

# SOMEET SAHOO

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## SUMMARY

Forward Deployed AI Engineer (Founding Team) specializing in **Agentic AI**, **RAG pipelines**, and **Voice/Audio AI** systems. Experience owning the end-to-end ML lifecycle—from scoping workflows with enterprise customers to deploying low-latency LLM agents in production. Expert in **Python**, **SQL**, and orchestrating multi-agent frameworks (**LangGraph**) that automate complex, real-world business operations with high reliability.

## EXPERIENCE

**Consuma.ai** | Forward Deployed AI Engineer (Founding Team) [\[System Architecture\]](#) **Bengaluru, India** | **Dec 2024 – Present**

- **Agentic AI & Reasoning:** Architected a multimodal validation pipeline (Video, Image, Text) processing daily submissions for **10,000+ schools**. Designed a Multi-Agent architecture using **LangGraph** and **LangChain**, implementing advanced **Chain of Thought (CoT)** and **Tree of Thoughts (ToT)** prompting to handle unstructured student data autonomously.
- **Data Engineering & Visibility:** Managed end-to-end data storage using **MySQL** and **MongoDB**. Engineered robust analytics pipelines via **GCP BigQuery** to power **Looker Studio** dashboards, delivering real-time program monitoring metrics directly to government stakeholders and Operations teams.
- **Cloud Infrastructure & Backend:** Built scalable backend microservices using **FastAPI** and **Docker**. Orchestrated and deployed production AI workloads across **GCP VMs** and **Cloud Run** to ensure high availability and sub-second latency.
- **Voice AI & Cost Optimization:** Led a Voice AI initiative to optimize multi-lingual transcription requests. Engineered highly accurate speaker diarization pipelines using **Pyannote-audio**, drastically reducing API inference costs for complex, real-world audio data.

**Indian Institute of Technology, Goa** | Research Intern (Deep Learning) **Goa, India** | **July 2023 – Nov 2023**

- Developed custom **Physics-Informed Neural Networks (PINNs)** from scratch using **PyTorch** to solve non-linear PDEs, manually optimizing complex loss landscapes and custom gradient flows.
- Benchmarked novel model architectures against traditional numerical solvers, reducing computational overhead while maintaining high-fidelity accuracy.

**Indian Oil Corporation Ltd.** | Data Analyst Intern **Odisha, India** | **Dec 2022 – Feb 2023**

- Built **ETL pipelines** using **Python** and **SQL** to process high-frequency sensor data, establishing the structured data layer for downstream predictive analytics.
- Developed predictive equipment-monitoring models achieving **>93%** accuracy by engineering robust features to handle noise and outliers in raw industrial sensor telemetry.

## PROJECTS

**Scribe: Multi-Agent Audio Transcription Platform** [\[System Design\]](#) *(Voice AI & Full-Stack Architecture)*

- Architected a full-stack platform using **FastAPI**, **React**, and **Supabase** (SQL), orchestrating a multi-agent system to process and transcribe large media files via **Google Gemini**.
- Designed a collaborative workflow: "Worker Agents" transcribed audio chunks concurrently, while a "Supervisor Agent" reconciled speaker diarization across chunk boundaries using rolling context windows.
- Implemented client-side media compression using **FFmpeg WASM** and engineered an asynchronous **FastAPI** backend with WebSockets for real-time transcription progress streaming.

**Enterprise RAG & Workflow Automation Agent** *(LLM Orchestration)*

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and **Pinecone** (Vector DB) to ingest and query unstructured technical and business documents.
- Implemented advanced context engineering and dynamic document chunking strategies to minimize hallucinations and ensure grounded responses for enterprise users.

**Edge-Optimized Inference Engine** *(Latency & Performance Tuning)*

- Designed a custom computer vision model (CNN), applying **Quantization (INT8)** and **Pruning** to compress the model and reduce inference latency by **70%**.
- Deployed compressed models onto resource-constrained edge hardware, maintaining **>92%** precision while achieving real-time execution speeds.

## SKILLS

- **AI & Frameworks:** LangGraph, LangChain, RAG, Voice AI (Gemini, Whisper, Pyannote), Multi-Agent Systems, Prompt Engineering (CoT, ToT), LLM Evaluation.
- **Backend & Cloud:** Python, FastAPI, Docker, GCP (Cloud Run, VMs, BigQuery), AWS, Supabase.
- **Databases & Viz:** PostgreSQL, MySQL, MongoDB, Vector Databases (Pinecone, Chroma), Looker Studio.
- **Domain & Ops:** End-to-End ML Lifecycle, Customer-Facing Engineering, Latency Optimization.

## EDUCATION

- **Institute of Chemical Technology, Mumbai** – BTech in Chemical Engineering **CGPA: 8.73**